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# Bi-directional ferrite to austenite transformation through warm temperature deformation of a ferrite-based low density steel

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ABSTRACT

The Gibbs energy model as an effective thermodynamic function along with detailed microstructural analyses were employed to assess the capability of strain induced transformations in a ferrite based lightweight steel. In this respect, the material was deformed in a warm temperature regime in compression mode and the resultant microstructures were studied in details. The progressive and extensive development of the banded structure was led to the partitioning of the ferrite phase and also the shearing of the parent austenite. The shear strain localization was effectively released through simultaneous transformation of ferrite to austenite and austenite to ferrite transformations as an effective strain compensation mechanism. This was led to appreciable flow softening in corresponding flow curves. The capability for bidirectional transformation resulted from the specified local elemental partitioning of Al and Mn and the resultant exactly same levels of Gibbs energies of each constituent phases.

## 1. Introduction

Various deformation mechanisms: (*i*) cell-structure formation and substructure development [1], (*ii*) short-range ordering and formation of Taylor lattice structure [2], (*iii*) micro-shear band induced plasticity [3,4], and (*iv*) slip band refinement and grain subdivision [2,5], have been proposed to justify the high capability of low density steels for dynamic strain hardening. These specified micro-mechanisms are exceptional ways for strain compensation which finally results in outstanding strength/ductility balances [6,7]. Interestingly, the nano-precipitation of kappa phase has been also introduced as an effective strain accommodation mechanism [8].

Another aspect which has been mainly overlooked in previous researches is the capability of lightweight steels for the occurrence of dynamic transformation in the course of straining. Owing to the high stacking fault energy of low density steels (80–110 mJ/m$^2$), the austenite to martensite transformation is theoretically impossible and the transformation induced plasticity effect is disregarded, however, the probability of deformation induced austenite to ferrite and ferrite to austenite transformations should be re-considered. In this respect, the thermodynamic interpretation of the microstructural evolution is introduced as an effective solution, and the "modified Gibbs energy model" can be successfully employed to interpret the kinetic of phase transformation in multicomponent alloying systems [9]. Such approach has been previously applied in the case of low carbon steels [10] and advanced high strength steels [11], just for confirming the experimental results [12].

In the case of lightweight steels, the microstructural evolutions seem to be more complex, and the effects of alloying element and their partitioning effects on the phase stability and stress/strain induced transformations [13], phase fraction and phase distribution [14] are more prominent. In this respect, the role of thermodynamic interpretation combined with detailed microstructural analysis should be overemphasized; this is highly necessitated for successful alloy designing [15] and results in a compromise between the manufacturing costs and quality of the final product. In the present work, such approach has been

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employed to justify the characteristics of strain induced transformations in a triplex (Ferrite-Austenite-Martensite) lightweight steel. This concept has been mainly overlooked in previous researches especially in the case of austenite to ferrite (and vice versa) transformation.

## 2. Material and methods

Fe-11.15Mn-5.6Al-0.07C (wt.%) low density steel was received in an electroslag remelted (ESR) condition. ESRed material was homogenized at 1200 °C for 1 h due to the elimination of any probable segregation in the microstructure, then hot-rolled through 5 passes in the temperature range of 1100–1200 °C followed by air quenching, and finally annealed at 950 °C for 1 h for attaining a triplex (ferrite + austenite + martensite) microstructure. The detailed characteristics of the annealed microstructure have been described in our previous research [16]. The high temperature compression tests were carried out by Gotech AI-7000 universal testing machine equipped with a programmable furnace in a wide range of temperature (100–700 °C) under the strain rate of 0.001 $s^{-1}$ up to the true strain 0.6. The specimens were then immediately water quenched to preserve of deformed microstructure at high temperatures. The compression specimens were soaked at deformation temperature for 5 min before testing to reach the thermal equilibrium condition. Microstructures were investigated with the aim of (*i*) field-emission scanning electron microscope (FE-SEM CZ/MIRA I LMH, TESCAN, Czech) equipped with an electron backscattered diffraction (EBSD) detector (Hikari, AMETEK EDAX, U.S.A) using a step size of 200 nm and (*ii*) transmission electron microscopy (TEM: JEM-2100-F with acceleration voltage: 200 KV equipped field-emission gun, developed by JEOL Ltd, Japan). Thin-foils for TEM analysis were prepared by twin-jet polishing using a solution of HClO4 (10%) and methyl alcohol (90%) at −30 °C and an applied potential of 16V. The equilibrium percentage of each presence element such as Mn and Al in the constituent ferrite and austenite phases and also the Gibbs free energy (GFE) was calculated through Thermo-Calc software utilizing the TCFE9 database.

## 3. Results and discussion

The triplex starting microstructure of the experimented material consists of ferrite, austenite and thermally induced martensite is given in Fig. 1a–b. The diffraction patterns of different selected areas clearly verify the existence of martensite which has been formed in the course of the previous annealing. The band slope phase contrast as a qualitative approach has been employed to differentiate between the martensite and ferrite phases both holding body centered cubic (BCC) crystal structure. The obtained results are depicted in Fig. 1a. The same phase identification approach has been successfully employed previously [16–18].

The initial fresh microstructure is characterized by the extremely low dislocation density of ferrite and austenite phases. This specified microstructure has been compressed down to a true strain of 0.6 in a wide range of temperature (100–700 °C) the results of which are summarized in Fig. 1c. The mechanical stability over the temperature range of 200–500 °C is the main characteristic of the flow curve as that is clearly evident in Fig. 1d. However, a significant drop in stress level is recognized with increasing the temperature from 500 to 600 °C (follow the blue-dashed oval). An appreciable flow softening after the true strain of 0.3 is recognized in compressive flow curves which have been obtained at 500–600 °C. Considering the low probability of dynamic recrystallization and extended dynamic recovery at these low temperature regimes, the observed flow softening can be theoretically attributed to the occurrence of strain induced transformations.

In this respect, the deformed microstructures are examined at interrupted strains of 0.3 and 0.6. Interestingly, the thermomechanical processing at the lower temperature of 500 °C has resulted in the frequently distributed deformation bands through the microstructure (Fig. 2a), the feature that has not been observed at 600 °C (Fig. 2b) due to the higher capability for strain distribution at higher temperature. In addition, corresponded phase maps of those two mentioned circumstances prove the triplex microstructure after deformation up to 0.6 strain (see Fig. 2c and d). In a more detailed view, at the interrupted strain of 0.3, many dislocations were tangled to form the cell structures within the ferrite (Fig. 3a), and a planar slip-like character has started to form.

The strain within the ferrite is mainly accommodated through activation of multiple slip systems: {110} <111> and {112} <111>, therefore, formation and intersection of slip bands is highly probable (follow the single-slip traces in Fig. 3a). In fact, the wavy character of dislocation movement and high dislocation maneuverability provide a proper condition for the formation of dislocation cells. In this respect, high dislocation density walls (HDDWs) can be traced through the microstructure. Accordingly, the development of banded structure within the ferrite phase at this relatively low temperature seems to be reasonable. The dislocation configuration in austenite was quite different at the same imposed strain of 0.3, as it is observed in Fig. 3b. The white arrows and correlated diffraction patterns indicate the formation of fresh ferrite grains within the prior austenite. The strain induced austenite to ferrite transformation, in the present case, can be considered as strain compensation mechanisms and justify the observed flow softening behaviour at such low temperature regime (Fig. 1c).

The higher magnification of the microstructures which have been compressed down at 500 °C to true strain of 0.6 are brought in Fig. 3c and d, where the banded structure is clearly revealed by dashed-black

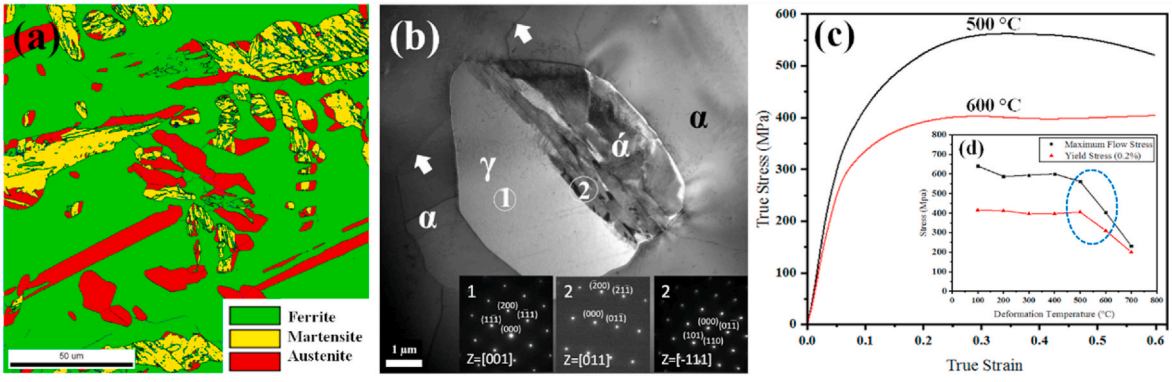



**Fig. 1.** Initial microstructure of triplex (ferrite-austenite-martensite) low density steel which has been annealed at 950 °C/1h: (a) EBSD Phase Map (PM) and (b) TEM bright-field micrograph and corresponding Selected Area Diffraction (SAED) patterns (white arrows shows the initial ferrite-ferrite boundaries). (c) True-stress-strain curves of the material which has been compressed under the strain rate of 0.001 $s^{-1}$ at 500 and 600 °C, and (d) extracted stress levels vs. temperature in the wide range of 100–700 °C.

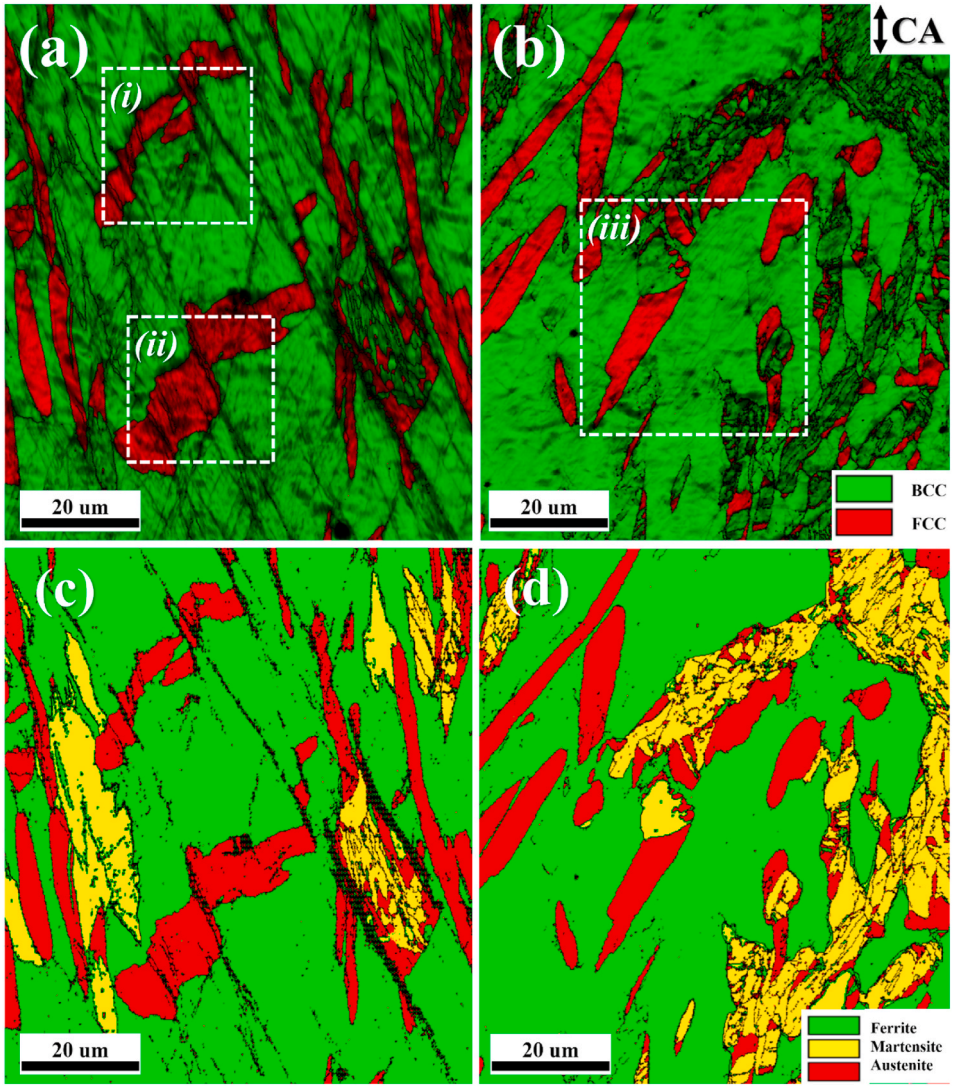



**Fig. 2.** EBSD PM overlaid on image quality (IQ) maps of experimented low density steel compressed at (a) 500 and (b) 600 °C under the strain rate of 0.001 $s^{-1}$ down to the true strain of 0.6. (c) and (d) are corresponded PMs which separate three constituent phases from each other. CA: Compression Axis.

ovals. As it can be realized, the parent austenite has been effectively sheared through the propagation of the deformation bands. Interestingly, the simultaneous occurrence of the deformation induced "ferrite to austenite" and "austenite to ferrite" transformations are recognized along with the deformation bands which are depicted by red and green arrows, respectively. In fact, the shear strain localization along the deformation bands is released through the transformation of the austenite to ferrite or ferrite to austenite as an effective emerging strain compensation mechanism. Such capability of the material for transformation-based strain accommodation reduces the flow localization's effect, causes flow softening and increases the workability of the experimented material. The occurrence of deformation induced ferrite transformation has been previously reported in the case of Fe–Mn–Al–C alloys at higher temperatures where the probability of such transformation is theoretically reasonable [3]. However, in the present case, the simultaneous occurrence of deformation-induced ferrite transformation (DIFT) and deformation-induced austenite transformation (DIAT) at such a low temperature regime seems phenomenal and needs to be well clarified. It would be necessary to mention that such transformations can't be characterized in the microstructures which have been deformed at higher temperature of 600 °C (see Fig. 3e).

In this respect, the partitioning of Mn and Al elements as the function of temperature in the constituent phases are represented in Fig. 4a–b. Different phases are predicted by Thermo-Calc in both diagrams: (1) BCC_A2 and (2) FCC_A1 (Face Centered Cubic) represent the ferrite and austenite phase in order. Furthermore, (3) BCC_A1#2 is a phase with the BCC structure which owns an extremely high amount of Mn. Additionally, both Mn and Al partitioned into ferrite and austenite. As it is expected, Mn has been partitioned more in austenite and also Al in ferrite. The elemental map constructed based on energy dispersive spectroscopy analysis of the deformed microstructure at 500 °C, shows almost similar partitioning (see and compare Table 1 and Fig. 4a–b).

The variation of Gibbs Free Energy vs. Mn and Al content for both ferrite (BCC) and austenite (FCC) are depicted in Fig. 4c–f. Apparently, the lower Gibbs energy exposes the higher phase stability in an equilibrium state. Point "A" in each condition reveals the chemical composition at which both BCC and FCC possess the same Gibbs free energy and co-occurrence of them in the microstructure is probable. Applying the nominal chemical composition of material indicates the differences of Gibbs free energies for ferrite and austenite at 500 °C (Fig. 4c). Surprisingly, considering the elemental partitioning according to Fig. 4a and b and Table 1, the same Gibbs free energies are obtained for both ferrite and austenite (follow the dashed line in Fig. 4c). Accordingly, the provided driving force owing to the strain localization at deformation bands (Fig. 2a, c) can be effective, in the present case, for the simultaneous occurrence of deformation induced ferrite to austenite and

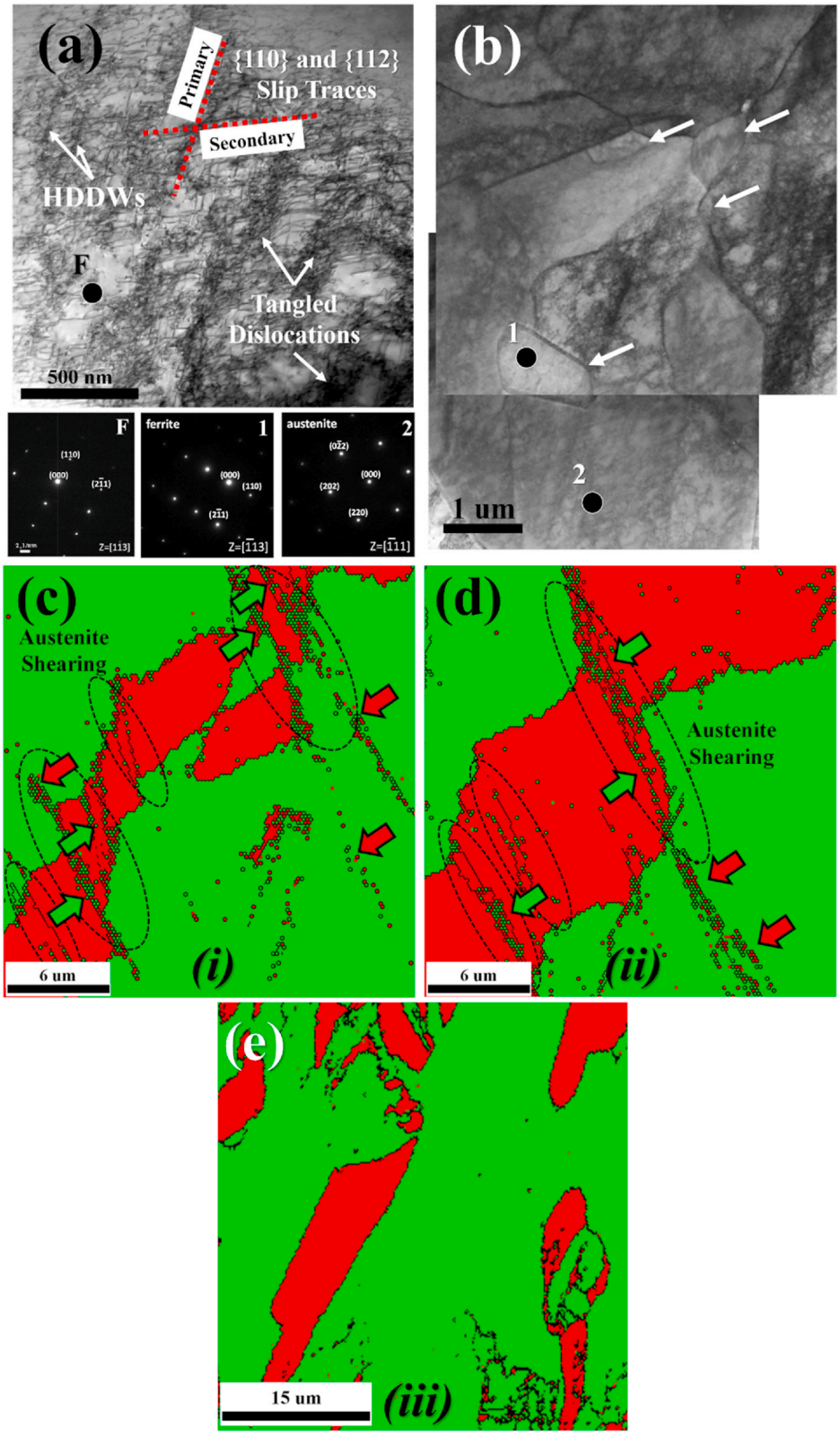



**Fig. 3.** TEM bright-field image of deformed microstructure at interrupted strain of 0.3 and 500 °C: (a) dislocation configuration and multiple slip traces within the ferrite phase and (b) formation of fresh ferrite grains (indicated by white arrows) within the parent austenite islands and the correlated diffraction patterns. (c–e) Represent the higher magnification of selected areas in Fig. 2a and b. Red and green arrows indicate the austenite and ferrite formation through deformation bands, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

austenite to ferrite transformations. On the contrary, almost the same Gibbs free energy for BCC and FCC have been obtained in the wide range of Al% (Fig. 4d and f), and changes in Al content does not have any remarkable influence on the phase stability at various conditions. All in all, Mn partitioning mainly dictates the amount of Gibbs free energy and determines the phase stability of the material.

## 4. Conclusion

The bi-directional “ferrite to austenite” and “austenite to ferrite” deformation induced transformations were analyzed in Fe-11.15Mn-5.6Al-0.07C (wt.%) low density steel employing detailed microstructural analysis and through a proper thermodynamic Gibbs energy

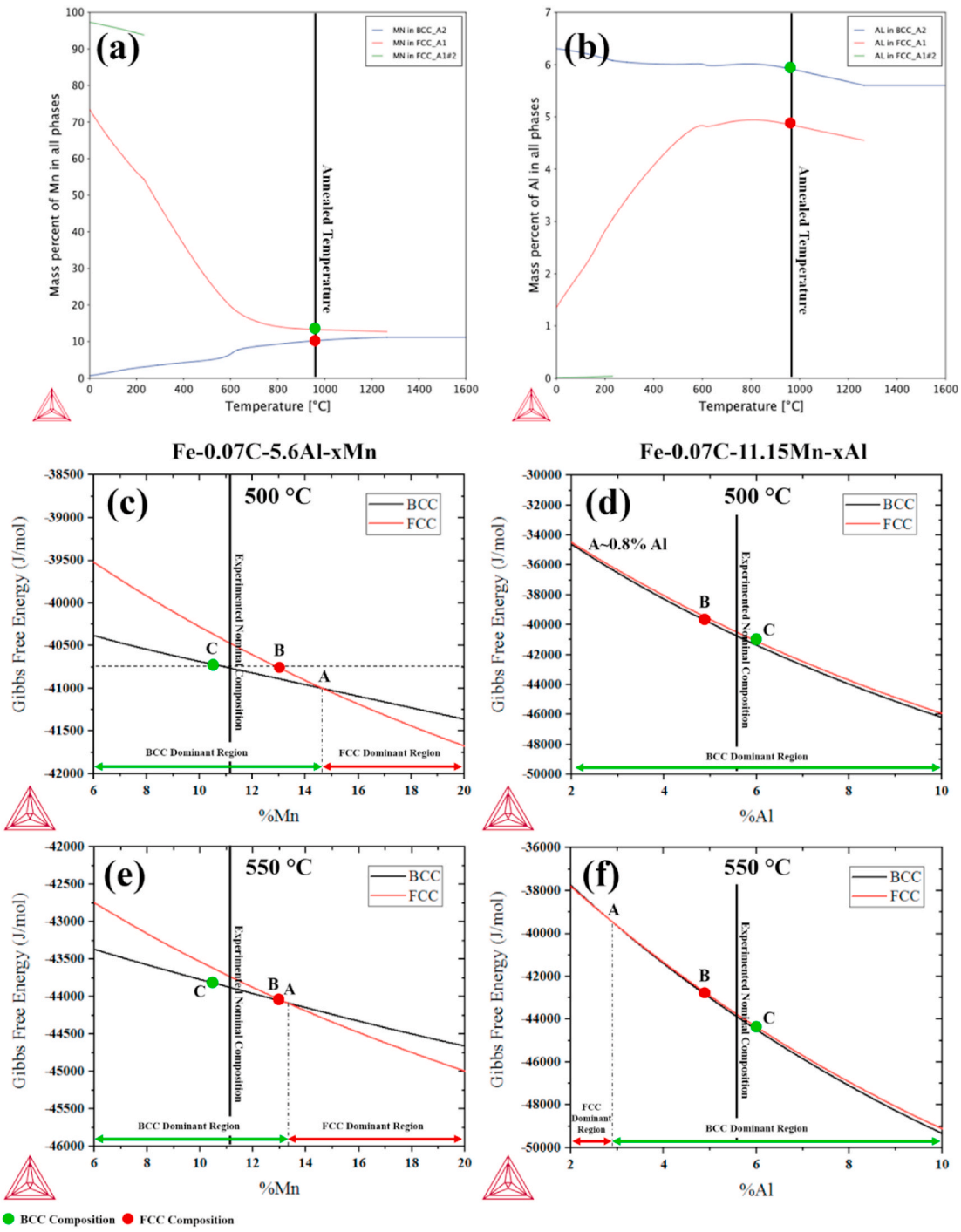



**Fig. 4.** The Thermo-Calc® results show the mass percentages of (a) Mn and (b) Al in each constituent phases in the equilibrium condition for experimented Fe-11.15Mn-5.6Al-0.07C system. The variation of Gibbs Free Energy of FCC and BCC at isothermal section of 500 and 550 °C with various content of (c, e) Mn% and (d, f) Al%. A: shows the point that Gibbs Free Energy of FCC and BCC is equal thermodynamically. B and C: show the FCC and BCC chemical composition due to the elemental partitioning which have been extracted from (a and b) and also from Table 1.

**Table 1**
Chemical composition of experimented steel after deformation at 500 °C–0.001 $s^{-1}$ up to the true strain of 0.6 from the SEM analysis.

| Element | Mn | Al |
|---|---|---|
| In ferrite (wt.%) | 10.3 ± 0.1 | 6.0 ± 0.1 |
| In Austenite (wt.%) | 12.8 ± 0.1 | 4.7 ± 0.1 |

model. In this respect, the thermomechanical processing was conducted in a wide temperature range of 100–700 °C. At warm temperature regime, many dislocations were tangled to form the cell structures within the ferrite and a planar slip-like character was formed. The frequency of slip bands was noticeable and formation/intersection of them was led to development of banded structure. The parent austenite was also effectively sheared through the propagation of these banded structure. Shear strain localization along the deformation bands was released through the simultaneous transformation of the austenite to

ferrite or ferrite to austenite as an effective emerging strain compensation mechanism. This resulted in reduction of flow localization's effect, caused flow softening and increased the workability of the experimented material. Bidirectional transformation was justified from thermodynamic point of view considering the exactly same level of Gibbs energy for both ferrite and austenite as a consequence of Al/Mn elemental partitioning.

## CRediT authorship contribution statement

**Amir-Reza Kalantari:** Conceptualization, Methodology, Investigation, Resources, Software, Validation, Writing – original draft. **Abbas Zarei-Hanzaki:** Supervision. **Hamid Reza Abedi:** Advisor, Methodology, Investigation, Validation, Writing – review & editing. **Seong-Jun Park:** Characterization, Writing – review & editing. **Jun Young Park:** Characterization, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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